BATTERY AGGREGATOR PRE-VISIT HOMEWORK

A practical study guide before meeting registered battery recyclers

Your objective
Understand enough regulation, chemistry, pricing, logistics and commercial terms to avoid buying the wrong batteries at the wrong price. Your first business is aggregation and reverse logistics, not recycling.

One-sentence introduction to use:
“I am building an organized waste-battery aggregation and reverse-logistics network, initially sourcing segregated end-of-life batteries from service centres, battery dealers, scrap aggregators, EV workshops and institutional generators, and supplying them only to appropriately registered recyclers.”

Study plan

Session	Topic	Time
1	Government framework and legal boundaries	35 min
2	Battery chemistries and material value	45 min
3	Pricing, deductions and per-kilogram economics	35 min
4	Logistics, safety and documents	25 min
5	Rehearse the recycler questions and introduction	30 min

1. Government framework you must know

The principal framework in India is the Battery Waste Management Rules, 2022. It covers waste-battery collection, segregation, transport, refurbishment and recycling across battery types. Producers, recyclers and refurbishers use the centralized battery Extended Producer Responsibility portal.

Abbreviation	Full form	Why it matters
MoEFCC	Ministry of Environment, Forest and Climate Change	Central ministry that notified the rules
CPCB	Central Pollution Control Board	National regulator and battery EPR portal operator
SPCB	State Pollution Control Board	State environmental regulator
KSPCB	Karnataka State Pollution Control Board	Relevant for Karnataka operations
APPCB	Andhra Pradesh Pollution Control Board	Relevant for Andhra Pradesh operations
BWM Rules	Battery Waste Management Rules, 2022	Principal battery-waste framework
EPR	Extended Producer Responsibility	Producer responsibility for end-of-life batteries
CTE / CFE	Consent to Establish / Consent for Establishment	Approval before establishing an applicable facility
CTO / CFO	Consent to Operate / Consent for Operation	Approval before operating an applicable facility
HOWM Rules	Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016	Relevant to specified hazardous waste handling and transport
TSDF	Treatment, Storage and Disposal Facility	Authorized destination for specified hazardous residues
MSME	Micro, Small and Medium Enterprise	Likely business classification
GSTIN	Goods and Services Tax Identification Number	Tax identity for invoicing
PPE	Personal Protective Equipment	Worker safety equipment
SOP	Standard Operating Procedure	Documented operating method

Your legal boundary as an aggregator

- Your proposed role is to collect, identify, segregate, temporarily store and transport batteries to registered recyclers.
- Do not open, break, drain, discharge, dismantle, crush or shred batteries unless the exact activity and facility are expressly permitted.
- A normal trading registration alone should not be assumed to authorize a battery collection centre. Ask the recycler how its approved collection partners are structured, then confirm the arrangement with the applicable State Pollution Control Board.
- Keep lead-acid and lithium-ion streams physically and operationally separated.

Remember: Collection is not recycling. Your safest starting structure is a written collection/reverse-logistics arrangement with a registered recycler or producer, supported by regulator confirmation.

2. Stakeholders and material flow

Stakeholder	Role	Your relationship
Producer	Manufacturer or importer placing batteries on the market	Potential collection contract or EPR-linked client
Recycler	Processes waste batteries and recovers defined materials	Your principal offtake customer
Refurbisher	Repairs or restores batteries for further use	Possible counterparty for reusable batteries
Aggregator / collection partner	Receives and consolidates segregated battery waste	Your proposed starting role
Waste generator	Service centre, dealer, fleet, institution, repair shop or scrap supplier	Your feedstock source

EPR in simple language

Extended Producer Responsibility means that the producer is responsible for meeting recycling obligations for batteries introduced into the market. Registered recycling and sale of eligible recovered materials can generate EPR certificates. As an aggregator, do not assume that you own the EPR certificate. Contractually establish who receives the certificate and whether its value is already included in the recycler's purchase price.

Key commercial question: "Does your purchase price already include the economic value associated with the Extended Producer Responsibility certificate?"

Processing levels to recognize

Level	What happens	Typical output
Collection / aggregation	Receive, identify, segregate, store and dispatch	Whole batteries, packs, modules or cells
Mechanical pre-processing	Controlled discharge, dismantling, shredding and physical separation	Black mass, copper, aluminium, steel and plastics
Chemical refining	Leaching and purification of black mass	Lithium, nickel, cobalt and manganese compounds
Integrated recycling	Mechanical and chemical stages in one business	Recovered battery materials / salts

3. Battery construction: cell, module and pack

A recycler may quote different prices for complete packs, modules and loose cells. Never treat them as equivalent.

Term	Meaning	Commercial effect
Cell	Smallest electrochemical unit, such as cylindrical, prismatic or pouch cell	Higher concentration of electrochemical material than a complete pack
Module	Multiple cells assembled electrically and mechanically	Contains cells plus holders, wiring and structure
Pack	Modules or cells plus enclosure, Battery Management System, busbars, cooling and connectors	Contains substantial non-cell weight; price is not equal to loose-cell price
Black mass	Concentrated electrode-bearing powder after mechanical processing	Not pure lithium or cobalt; value depends on chemistry, assay and impurities

Basic lithium-ion construction

- Cathode: determines the major chemistry, such as LFP, NMC or LCO.
- Anode: commonly graphite, supported on copper foil.
- Cathode current collector: commonly aluminium foil.
- Electrolyte: carries lithium ions between electrodes.
- Separator: prevents direct electrical contact while allowing ion movement.
- Battery Management System: monitors voltage, current, temperature and safety in a pack.

Memory rule: Complete pack ≠ module ≠ cell ≠ black mass. Always ask which physical form a quoted ₹/kg applies to.

4. Battery chemistry study table

Code / full name	Typical sources	Main recycling value	Aggregator view
LAB - Lead-Acid Battery	Petrol motorcycles, cars, trucks, UPS, inverter, telecom	Lead; sulfuric-acid system	Strong volume and mature offtake; keep completely separate from LIB
LFP - Lithium Iron Phosphate	EV 2W/3W, buses, passenger EVs, solar and energy storage	Lithium, copper and aluminium; no nickel or cobalt	Growing volume but weaker intrinsic value; buy only against buyer quote
NMC - Lithium Nickel Manganese Cobalt Oxide	EVs, industrial and high-energy packs	Lithium, nickel, manganese, cobalt, copper and aluminium	High priority; strong recycling interest
LCO - Lithium Cobalt Oxide	Phones, laptops, tablets and consumer electronics	Cobalt and lithium	High priority; attractive in segregated electronic-battery streams
NCA - Lithium Nickel Cobalt Aluminium Oxide	Certain high-energy EV and industrial packs	Nickel, cobalt and lithium	High interest; verify identity before pricing
LMO - Lithium Manganese Oxide	Power tools, medical devices and legacy mobility	Lithium, manganese, copper and aluminium	Secondary stream; do not pay NMC pricing
Ni-Cd - Nickel-Cadmium	Legacy industrial, railway, aviation and emergency systems	Nickel and cadmium	Specialist buyer only; cadmium hazard
NiMH - Nickel-Metal Hydride	Older hybrid vehicles and rechargeable equipment	Nickel	Segregated specialist stream
Zinc-based	AA, AAA and primary household batteries	Zinc, manganese and iron	Low value per kg; service/EPR model preferable

5. Chemistry memory cards

Lead acid

Formula / system: Pb / PbO2 / H2SO4
Remember: Petrol-bike and inverter stream. Mature lead recycling. Never mix with lithium-ion.

LFP

Formula / system: LiFePO4
Remember: No nickel or cobalt. Growing EV/storage volume but price conservatively.

NMC

Formula / system: Lithium Nickel Manganese Cobalt Oxide
Remember: Valuable nickel and cobalt. High-priority EV chemistry.

LCO

Formula / system: LiCoO2
Remember: Cobalt-rich consumer-electronics chemistry. Target IT and electronics channels.

NCA

Formula / system: Lithium Nickel Cobalt Aluminium Oxide
Remember: Nickel/cobalt-bearing high-energy chemistry. Verify before purchase.

LMO

Formula / system: LiMn2O4
Remember: Manganese-rich; generally lower intrinsic value than NMC.

Ni-Cd

Formula / system: Nickel-Cadmium
Remember: Cadmium requires specialist handling and buyer-backed procurement.

NiMH

Formula / system: Nickel-Metal Hydride
Remember: Nickel-bearing stream found in older hybrid and industrial applications.

Five chemistries to memorize before the visit: LAB, LFP, NMC, LCO and NCA.

6. Pricing and per-kilogram economics

Your profit is determined before you buy. Never purchase a battery lot first and search for a recycler later.

Maximum supplier buying price = recycler confirmed value - inbound collection - outbound freight - handling/storage - finance/admin - rejection allowance - required profit.

Worked example	₹ per kg
Recycler confirmed NMC purchase value	165
Collection and transport	-7
Handling and administration	-4
Chemistry / rejection risk allowance	-5
Required contribution	-20
Maximum supplier buying price	**129**

Why small price errors matter

At 10 tonnes per month, a ₹3/kg error costs ₹30,000 per month. At 30 tonnes, the same error costs ₹90,000 per month. Therefore, written rate confirmation, price-validity period, deductions and final weighing point matter more than headline turnover.

Never accept an unclear quote

- Ask whether the rate applies to complete packs, modules, cells, production scrap or black mass.
- Ask whether the price is delivered-to-plant or includes pickup.
- Ask how long the price is valid.
- Ask exactly how chemistry, moisture, casing and contamination are deducted.
- Ask whether payment is based on received weight, accepted weight, assay or recovered output.

7. Inspection and identification basics

Check	What to record	Why
Source application	Petrol bike, EV, laptop, UPS, inverter, telecom or industrial equipment	Application narrows likely chemistry
Brand and model	Label photo, serial number and model number	Supports identification and ownership trail
Physical form	Pack, module, cell or mixed material	Determines recycler price basis
Condition	Intact, swollen, leaking, burnt, opened or accident-damaged	Changes safety, acceptance and transport requirements
Chemistry evidence	OEM label, model database, recycler test or laboratory result	Prevents LFP/NMC mispricing
Weight	Gross, packaging and net material weight	Controls settlement dispute
Ownership	Invoice, declaration, auction document or producer authorization	Important for warranty and EV batteries

Do not rely only on appearance. Similar-looking cells may contain different chemistries. Ask the recycler what evidence it accepts and how it verifies LFP versus NMC.

Important technical terms

Term	Full form / meaning
BMS	Battery Management System
SOC	State of Charge: remaining charge level
SOH	State of Health: remaining usable condition/capacity
XRF	X-Ray Fluorescence: elemental analysis tool
Assay	Laboratory determination of material composition
Payable	Percentage of contained metal value recognized in commercial settlement
Treatment charge	Processing charge deducted from recoverable material value

8. Logistics, storage and safety homework

- Keep lead-acid, lithium-ion and specialist chemistries in separate identified areas and containers.
- Keep damaged, swollen, leaking or burnt lithium-ion batteries isolated from normal stock.
- Protect terminals against short circuit and prevent batteries from moving during transport.
- Use chemistry- and condition-appropriate packaging approved by the recycler and applicable transport requirements.
- Record source, date, chemistry, condition, weight, photographs and destination for each batch.
- Do not hold speculative inventory. Target rapid turnover and obtain recycler approval before collecting unusual or damaged material.
- Use trained personnel and suitable Personal Protective Equipment. Maintain emergency contacts and an incident procedure.

Documents to discuss with the recycler

Document	Purpose
Supplier declaration / purchase record	Shows source and legal transfer
Battery inventory / packing list	Records battery type, count, condition and weight
Photographs and label details	Pre-approval and chemistry identification
Tax invoice / purchase voucher	Commercial and tax trail
Transport document / manifest	Movement traceability where applicable
Weighment slip	Settlement evidence
Material receipt / recycling acknowledgement	Proof of delivery to registered recycler
Collection or offtake agreement	Defines allowed activities, prices, responsibility and EPR ownership

Before interstate movement: ask the recycler and both relevant State Pollution Control Boards what prior intimation, authorization or documentation applies to the exact battery category and route.

9. Commercial terms to understand before agreeing

Topic	What you must establish
Price basis	Battery chemistry and physical form to which ₹/kg applies
Price validity	Number of days the quoted rate is honoured
Minimum lot	Pickup and delivered minimum quantity
Freight	Who pays pickup and outbound transport
Weight basis	Source weight, plant-gate weight or accepted post-inspection weight
Deductions	Casing, plastic, moisture, contamination, damage and incorrect chemistry
Payment trigger	Receipt, inspection, assay or sale/recovery completion
Payment period	Same day, 2, 7, 15 or 30 days
Rejected loads	Return freight and safe handling responsibility
EPR ownership	Recycler, producer or contractually assigned party
Insurance / incident liability	Who bears fire, transit and environmental losses
Trial load	Quantity, chemistry, documentation and acceptance process

Preferred starting arrangement

Seek a written collection or reverse-logistics arrangement with at least two registered recyclers. Ask for a pre-purchase rate confirmation, approved packaging instructions, the required source documents and a small trial-load process. Avoid exclusivity until pricing, payment and rejection behavior are proven.

10. Red flags and walk-away conditions

1. Recycler will not show or provide an official path to verify current registration and operating consent.
2. One price is offered for “all lithium batteries” without chemistry or condition distinction.
3. Deductions and settlement weights are not explained in writing.
4. Cash-only transaction with no invoice, weighment slip or material receipt.
5. Recycler asks you to mix lead-acid and lithium-ion batteries in one load.
6. Recycler encourages unapproved dismantling, acid draining or pack opening.
7. Recycler asks you to create false source, ownership or chemistry documentation.
8. Price is far above every competing recycler but no commercial basis is provided.
9. Payment is dependent on an undefined future recovery result with no time limit.
10. You are expected to bear rejected-load return and safety costs without prior acceptance criteria.

Walk away if: registration cannot be verified, material ownership is unclear, chemistry cannot be established, or the expected contribution after all costs is below your minimum requirement.

11. What to observe during the plant visit

Area	Observation
Receiving	Is every load weighed, photographed and recorded?
Storage	Are chemistries and damaged batteries segregated?
Fire safety	Are quarantine, detection and suppression systems visible?
Traceability	Are lots labelled and linked to supplier/source records?
Processing	Does the actual equipment match the authorized activity claimed?
Housekeeping	Are batteries safely contained or piled randomly?
Worker safety	Are workers trained and using appropriate PPE?
Outputs	Can the recycler explain black mass, copper, aluminium, steel or recovered salts?
Documentation	Can it issue weighment, receipt and settlement records?

Do not photograph restricted areas without permission. The visit is to verify commercial and compliance capability, not to obtain confidential process information.

Three high-value discussion prompts

- Which battery material are you currently struggling to source?
- Which material do aggregators most frequently supply incorrectly?
- If you were starting a battery aggregation company today, which battery stream would you focus on and why?

12. Final revision page

Memorize these 12 points before entering the recycler’s office.

1. My current role is aggregation and reverse logistics, not dismantling or recycling.
2. I must verify CPCB/SPCB registration, authorization, consent validity and capacity.
3. Lead-acid and lithium-ion are separate supply chains.
4. LFP has no nickel/cobalt; NMC and LCO are generally more valuable to recyclers.
5. Pack, module, cell and black mass prices are different.
6. I will never buy without a recycler-confirmed price and acceptance criteria.
7. My maximum buy price includes all logistics, handling, finance, rejection and profit.
8. I need written deductions, final weighing point and payment trigger.
9. I must establish ownership and documents for warranty, accident and EV batteries.
10. I will ask who owns the EPR certificate and whether its value is in the price.
11. I will request a small trial load before committing large working capital.
12. I need at least two qualified recycler outlets for every important battery stream.

13. Self-test before the visit

1. What is the difference between a collection centre and a recycler?

2. What do CPCB, SPCB, EPR, CTE and CTO stand for?

3. Why should LFP never automatically receive the same buying price as NMC?

4. What is the difference between a cell, module, pack and black mass?

5. Which chemistry is common in petrol-bike starter batteries?

6. Which chemistry is often associated with phones and laptops?

7. What are the seven cost items deducted from recycler value to calculate your maximum buying price?

8. Who owns the EPR certificate in your proposed transaction?

9. What documents will accompany a trial load?

10. What will you do if the recycler refuses to provide written rejection criteria?

14. Visit-day checklist

☐ Print one Recycler Visit Question Sheet per recycler.
☐ Carry company introduction, identification and business contact card.
☐ Carry the list of battery streams you expect to source.
☐ Prepare three volume scenarios: 500 kg, 2 tonnes and 5 tonnes.
☐ Ask permission before visiting process or storage areas.
☐ Record registration number, consent validity and authorized capacity.
☐ Obtain chemistry-wise rates and specify pack/module/cell form.
☐ Record freight, deductions, final weight and payment days.
☐ Ask for collection-partner requirements and EPR treatment.
☐ Request a written trial-load process and procurement contact.
☐ Do not commit to a volume or exclusive agreement during the first meeting.
☐ Compare all recyclers using total net realization, not headline ₹/kg.

Your target outcome

Leave the visit with verified registration details, accepted chemistries, purchase prices by physical form, deductions, logistics costs, payment period, EPR treatment, aggregator-partner requirements and an agreed trial-load next step.

Important: This study guide is for commercial preparation. Confirm the final operating structure and permissions for your exact premises, activities, battery categories and transport routes with the applicable regulator and qualified compliance professional before handling waste batteries.